

Name: Yash Raj

Roll no: 24K-0737.

BLS-3H.

Question #1 :-

find Regular Expressions of the followings :-

- (a)  $(a+b)^3 + (a+b)^* (a+b)(a+b)(a+b)(a+b)^*$
- (b)  $(a+b)^* a (a+b)^* a (a+b)^*$
- (c)  $b^* (a+\lambda) b^* (a+\lambda) b^*$
- (d)  $(a+b)^* bb (a+b)^* bb (a+b)^*$
- (e)  $a^* + (a^* b a^* b a^*)^*$
- (f)  $((a^*) (a^* + b^* a a a^*)^* (b)^* (a)^*)$
- (g)  $a(a+b)^* b + b(a+b)^* a$
- (h)  $(ab+ba+aa) a^*$
- (i)  $(a+b)^* (aa+bb) (a+b)^*$
- (j)  $((aa+bb)^* \cdot (ab+ba) \cdot (aa+bb)^*) \cdot ((ab+ba) \cdot (aa+bb) \cdot (ab+ba))^*$

Question #2:- Write Regexp:-

- (a)  $^1 \backslash d \{2\} [A-Za-z] + \backslash + [A-Za-z] + @ \text{My-Email Service} . \text{com} \$$

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(b)  $\frac{1}{w} [A-Z] ( ? , \{14, \} \$ ) ( ? , * | d ) . ( ? , *$

$(A-Z0-9) [A-Z0-9 | d \{A-Z0-9\}^*$

(c)  $\wedge ( ? , * ( \text{comp} | \text{imp} ) ? ( ? , * 20 + ) ( ? , * \text{virus} \$ ) ^*$

? , look head



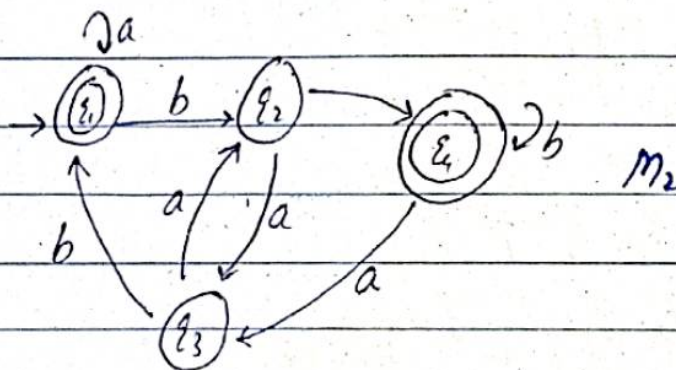
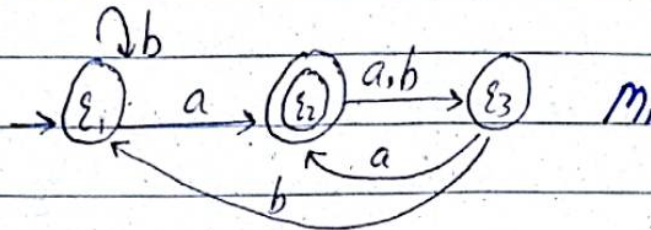
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BLS-3H

Section - 2 ::

Question #1:



(a) What is initial state:-

final state is the state where the automaton starts processing the input string.

→ usually denoted by a special arrow ( $\rightarrow \epsilon_0$ )

- In  $M_1$ ,  $\epsilon_1$  is the initial state.
- In  $M_2$ ,  $\epsilon_1$  is the initial state.

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(b) What is the set of acceptance states:-

Ans:- Acceptance states or also known as final state are the states in which the machine halts & accepts the input if it reaches them after processing all the input.

→ In  $M_1$ , there is only one final state or acceptance state which is  $q_2$ .

→ In  $M_2$ , there are two final state which includes  $q_1$  &  $q_4$ .

(c)  $M_1$

(i) aabb :  $q_2 q_2 q_1 q_1$

(ii) bbaa :  $q_1 q_1 q_2 q_2$

$M_2$

(i) aabb :  $q_1 q_1 q_2 q_4$

(ii) bbaa :  $q_2 q_4 q_3 q_2$

(d) (a)  $M_1$  : doesn't accept the "aabb" string.

(b)  $M_2$  : accepts the aabb string.

(e) (a)  $M_1$  : doesn't accept the  $\epsilon$ .

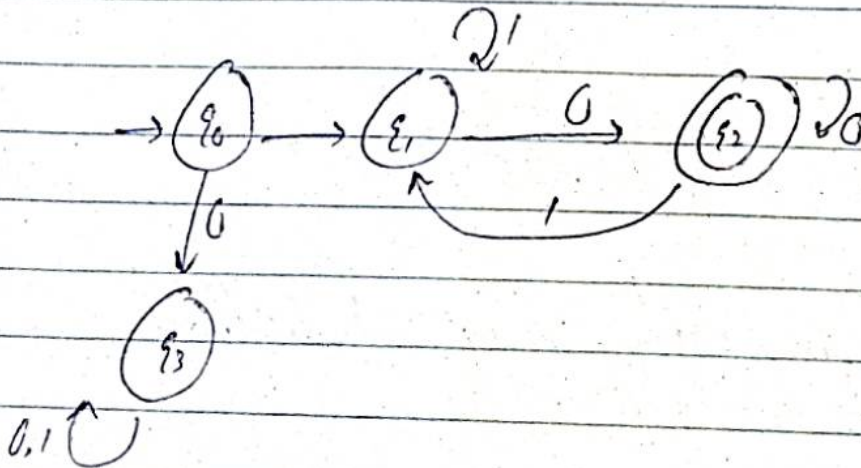
$M_2$  : accepts the  $\epsilon$ .



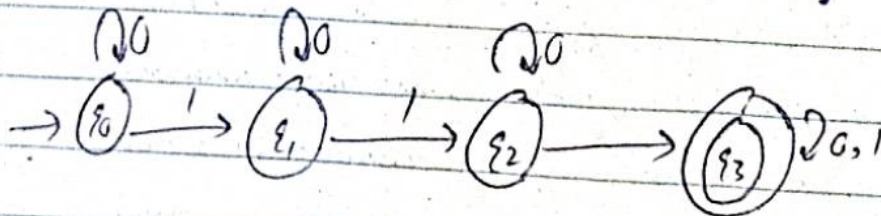
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Question #2:-

(a)  $\{w/w \text{ begins with a 1 \& ends with a 0}\}$



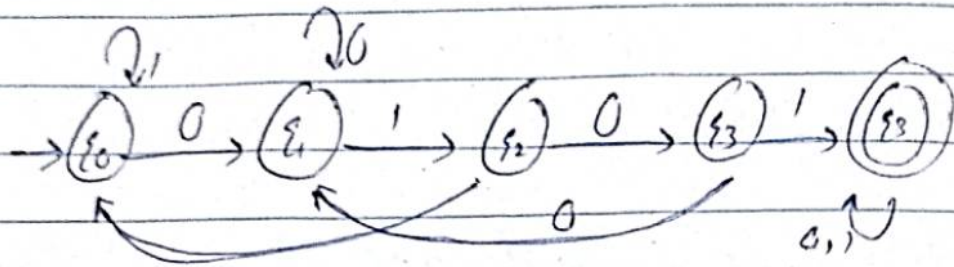
b)  $\{w/w \text{ contains at least three ss}\}$



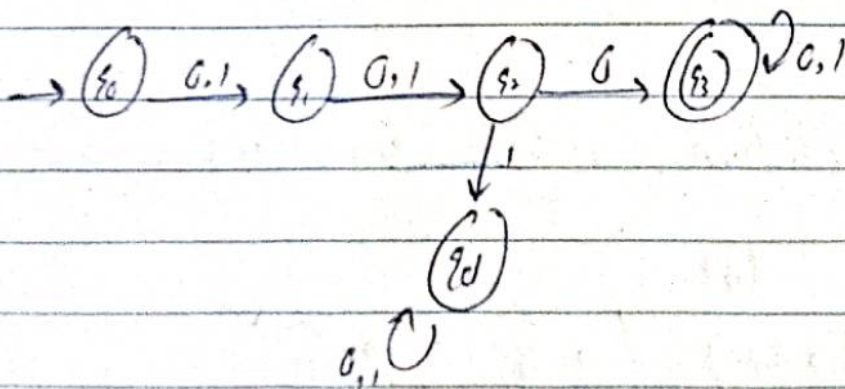
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(c) {w/w contains the substring 0101}



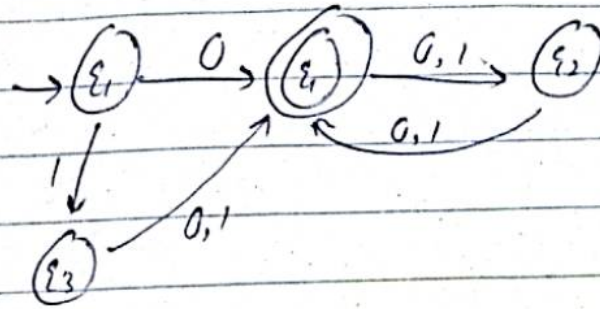
(d) {w/w has length atleast 3 & third symbol is a 0}



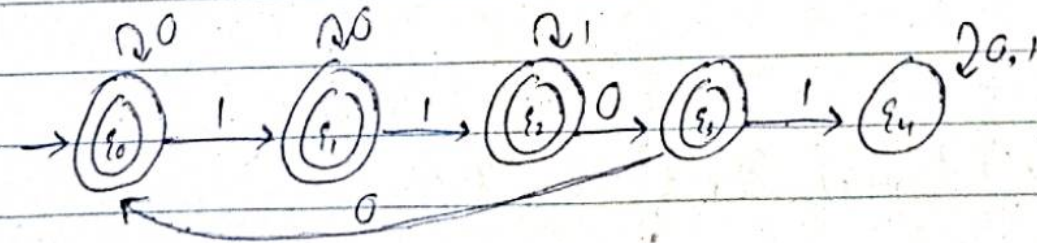
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(e)  $\{w/w \text{ starts with } 0 \text{ \& has length (odd), or starts with } 1 \text{ \& has even length}\}$



(f)  $\{w/w \text{ doesn't contain the substring } 1101\}$   
 $L = \{10010101, 10101, \dots\}$

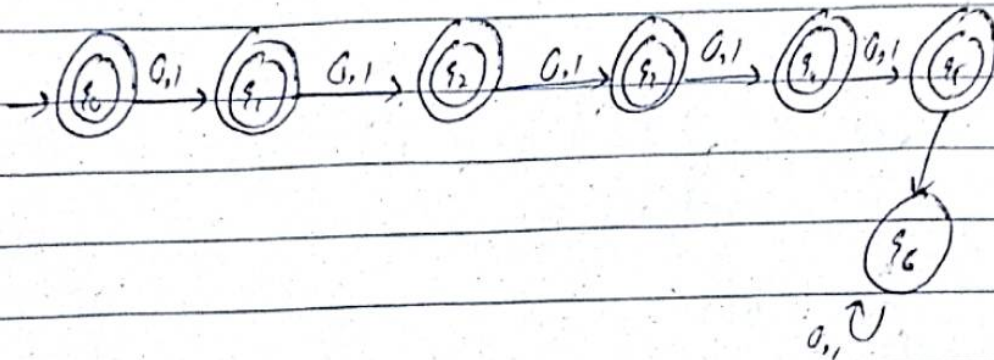




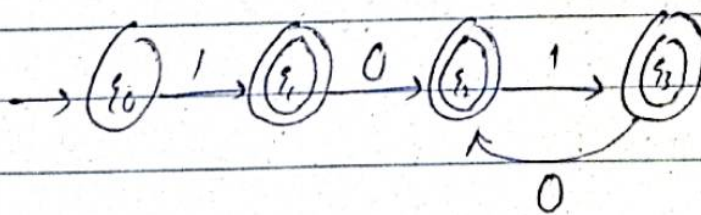
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(g)  $\{w \mid \text{the length of } w \text{ is at most } 5\}$   
 $L = \{\lambda, 0, 1, 00, 11, 0011, 10101, \dots\}$



(h)  $\{w \mid \text{every odd position of } w \text{ is a 1}\}$   
 $L = \{1, 101, 1010, 1010101, \dots\}$



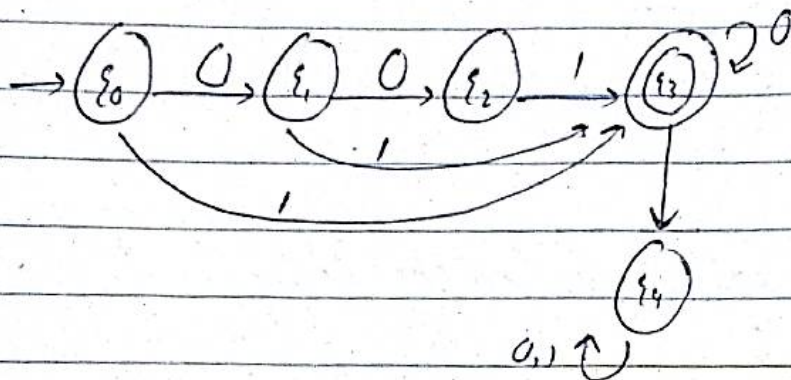


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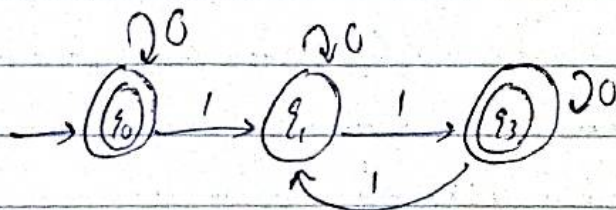
(i) {w/w contains atleast two 0s and atmost one 1}

$L = \{001, 010, 100, 01000 \dots\}$



(j) {w/w contains an even number of 1s, or contains exactly two 1s}

$L = \{\lambda, 11, 1101100110101 \dots\}$

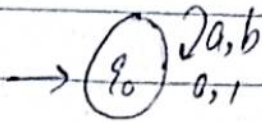


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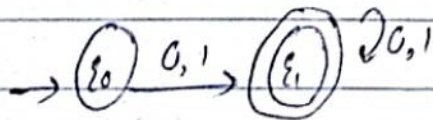
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K - Empty set.

$L = \{\}$

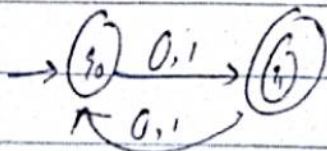


$L =$  All strings except the empty string.  
 $L = \{0, 1, 00, 101, \dots\}$



$M = \{w \mid \text{length of } w \text{ is odd}\}$

$L = \{1, 0, 001, \dots\}$

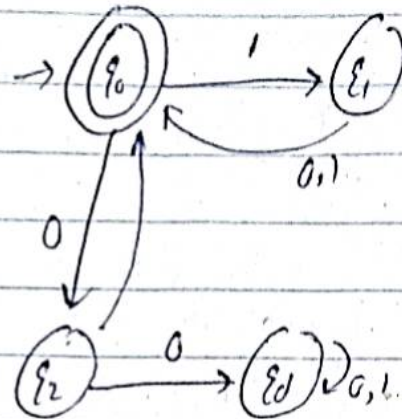




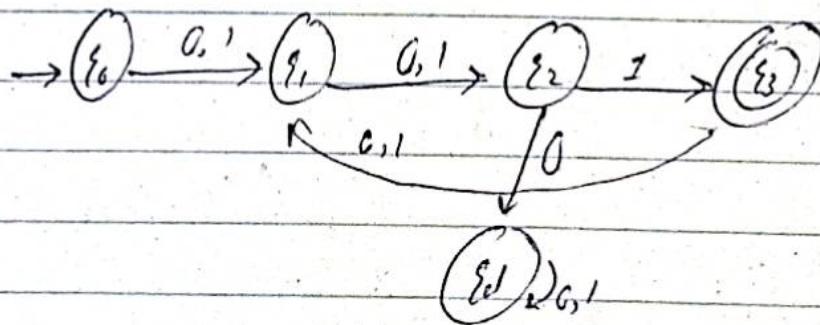
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$n \leq \{w / \text{length of } w \text{ is even \& it must contain } 00\}$



$0 \leq \{w / \text{every third character in } w \text{ is } 0\}$

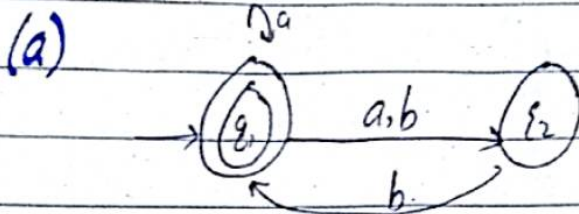


2) Section #3:

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Roll-no: 24K-0737

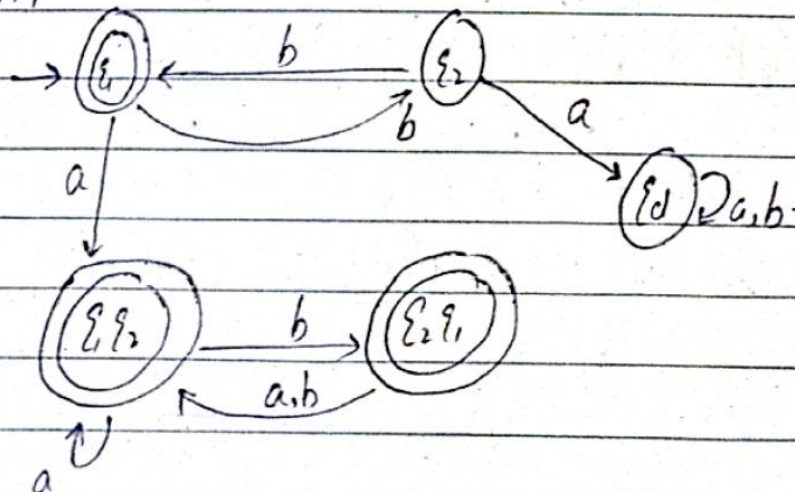
Question #1:



|                | a                             | b              |
|----------------|-------------------------------|----------------|
| q <sub>1</sub> | q <sub>1</sub> q <sub>2</sub> | q <sub>2</sub> |
| q <sub>2</sub> | -                             | q <sub>1</sub> |

|                               | a                             | b                             |
|-------------------------------|-------------------------------|-------------------------------|
| q <sub>1</sub>                | q <sub>1</sub> q <sub>2</sub> | q <sub>2</sub>                |
| q <sub>2</sub>                | Dead                          | q <sub>1</sub>                |
| q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub> q <sub>2</sub> | q <sub>2</sub> q <sub>1</sub> |
| q <sub>2</sub> q <sub>1</sub> | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub> q <sub>2</sub> |
| Dead                          | Dead                          | Dead                          |

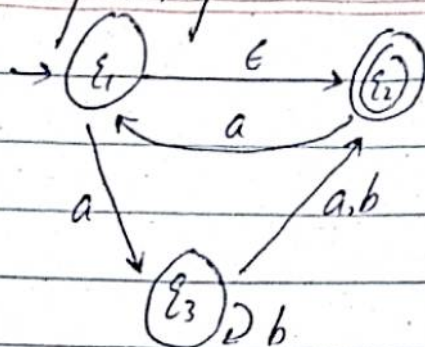
DFA:





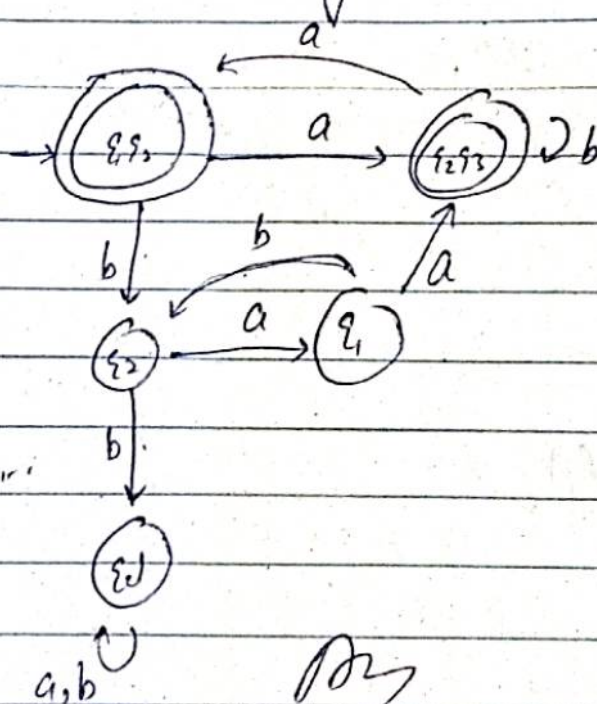
(b) Name: Yash Raj

Roll no: 24K-0737



Epsilon NFA to NFA

E(Epsilon) NFA to DFA by  $\epsilon^*$  (closure method)



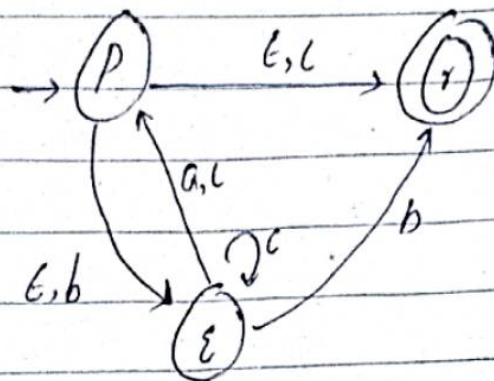
NFA to DFA

By

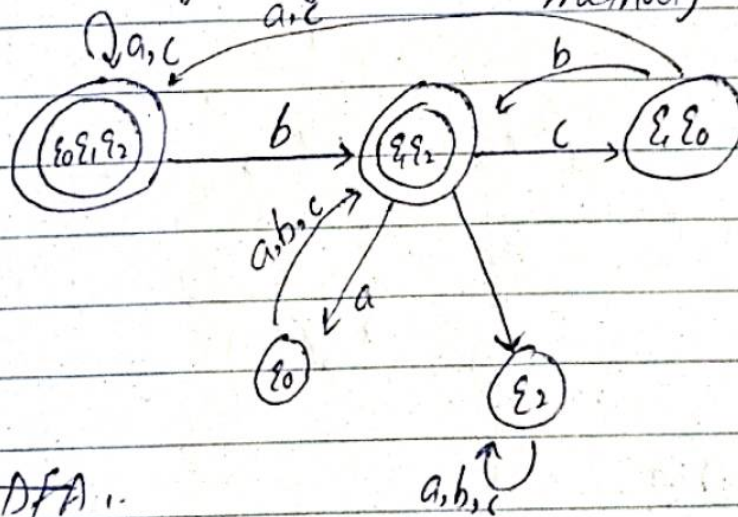
Name: Yash-Raj

Roll-no: 24K-0737

(1)



→ ε-NFA to DFA by using  $\epsilon^*$  (Epsilon-closure method)



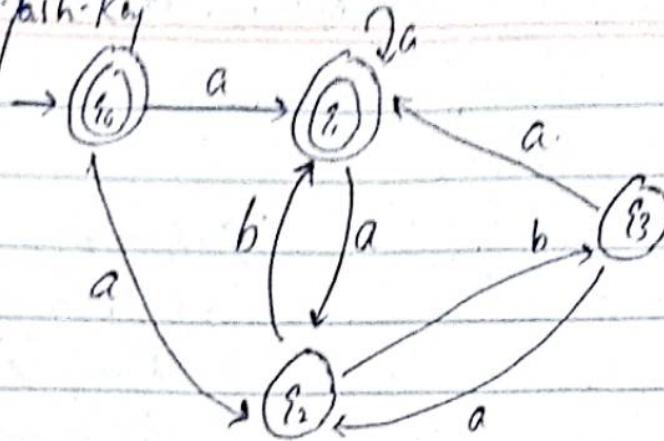
→ NFA to DFA



Name: Ash-Ray

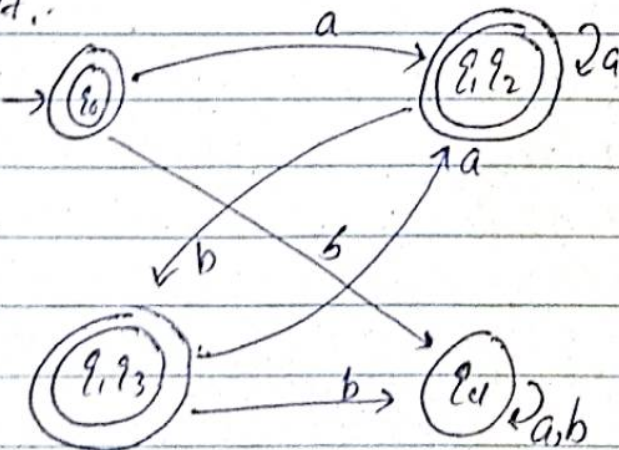
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(d)



|                | a                             | b                             |                               | a                             | b                             |
|----------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| q <sub>0</sub> | q <sub>1</sub> q <sub>2</sub> | -                             | q <sub>0</sub>                | q <sub>1</sub> q <sub>2</sub> | Dead                          |
| q <sub>1</sub> | q <sub>1</sub> q <sub>2</sub> | -                             | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub> q <sub>2</sub> | q <sub>1</sub> q <sub>3</sub> |
| q <sub>2</sub> | -                             | q <sub>1</sub> q <sub>3</sub> | q <sub>1</sub> q <sub>3</sub> | q <sub>1</sub> q <sub>2</sub> | Dead                          |
| q <sub>3</sub> | q <sub>1</sub> q <sub>2</sub> | -                             | Dead                          | Dead                          | Dead                          |

DFA:



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2) Section-4:-

States:-

- S<sub>1</sub> : Welcome screen
- S<sub>2</sub> : Validating card
- S<sub>3</sub> : 1st pin Entering screen
- S<sub>4</sub> : 2nd pin Entering screen
- S<sub>5</sub> : 3rd pin Entering screen
- S<sub>6</sub> : Choosing transaction screen
- S<sub>7</sub> : performing transaction
- S<sub>8</sub> : Asking for other transaction screen
- S<sub>9</sub> : Terminate ATM session

Output:-

- 0 : Nothing happens
- 1 : Eject ATM card
- 2 : Lock ATM card

Input:-

- A : Insert ATM card
- B : Valid ATM card
- C : Invalid ATM card
- D : Correct pin
- E : Incorrect pin
- F : Transaction is selected
- G : Complete Transaction
- H : Yes, for other transaction
- I : No, for other transaction



